JOYKIRAT SINGH

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EDUCATION

University of North Carolina at Chapel Hill — **Chapel Hill, NC**
*Ph.D. in Computer Science (**Advised by Prof. Mohit Bansal**)* — *2025 – Present*

Indraprastha Institute of Information Technology, Delhi — **New Delhi, India**
*B.Tech in Computer Science and Design (**Silver medalist**); **GPA: 9.24/10.0*** — *2019 – 2023*

RESEARCH INTERESTS

My research focuses on reinforcement learning for reasoning and agentic LLMs, building agents that reason efficiently and track their beliefs and uncertainty over long horizons. Currently, I focus on data generation and environment editing: creating environments that expose where agents are miscalibrated or missing key behaviors.

EXPERIENCE

MURGe-Lab Group — **Chapel Hill, NC**
Research Assistant (Advisor: Mohit Bansal) — *Aug 2025 – Present*
- Improved the reasoning abilities of language models while enhancing their efficiency through adaptive difficulty-based attention compression. (COLM 2026)
- Decoupled LLM agents into belief state and policy models jointly trained with RL, representing state uncertainty as verbalized certainty over atomic claims. (Preprint)

Google Cloud — **Sunnyvale, CA**
PhD Intern (Host: Sravan Bodapati) — *May 2026 – August 2026*
- RL-finetuned Gemini-3.5-Flash to reduce P90 latency by 30% on an NL-to-SQL agentic task while preserving task accuracy.
- Designed a hypothesis-and-verification loop that detects early error signals using error taxonomy classification in coding agent trajectories, enabling mid-trajectory correction and improving end-to-end task success.
- Developed an inference-time co-evolution framework where an agent and a verifier iteratively improve each other from past rollouts alone, with no ground-truth supervision or model weight updates.

Microsoft Research — **Bengaluru, India**
Research Fellow (Principal Researcher: Akshay Nambi) — *Jul 2024 – May 2025*
- Enhanced large language models' capabilities through agentic RL-based tool integration (NeurIPS FoRLM Workshop 2025).
- Investigated LLMs' capacity to detect and correct reasoning errors, revealing fundamental weaknesses in their step-by-step reasoning processes (ACL Main 2025).
- Developed an automated, feedback-driven framework for task-aware and efficient prompt optimization (ACL Finding 2025).
- Enhanced small language models' mathematical reasoning through dense-reward MCTS training and Direct Policy Optimization for step-wise improvement.

Laboratory for Computational Social Systems (LCS2), IIT Delhi — **New Delhi, India**
Graduate Research Assistant under Prof Tanmoy Chakraborty and Prof Soumen Chakrabarti — *Jun 2023 – Jun 2024*
- Revealed internal mechanisms driving step-by-step reasoning through mechanistic interpretability. (TMLR 2024)
- Improved arithmetic reasoning via symbolic solvers, low-rank adaptation, and RL (AAAI 2024).
- Developed a data-efficient task adaptation method that aligns model behavior through targeted neural rotations (EMNLP Finding 2025).

Expedia Group — **New Delhi, India**
Software Development Engineer, Payment's Module — *Aug 2023 – Jun 2024*
- Contributed to payment and conversation modules using Java/Kotlin, implementing CI pipelines, Datadog metrics, and LanguageIO-based translation for production systems.

Google Summer of Code — **Remote**
Intern, MusicBlock — *May - Aug 2021*
- Contributed to MusicBlock 2.0 using React, TypeScript, and P5, implementing MVVM-based palette and artboard features and improving UX from version 1.0.

PUBLICATIONS

Agent-BRACE: Decoupling Beliefs from Actions in Long-Horizon Tasks via Verbalized State Uncertainty
Joykirat Singh, *Zaid Khan, Archiki Prasad, Justin Chih-Yao Chen, Akshay Nambi, Hyunji Lee, Elias Stengel-Eskin, Mohit Bansal* (preprint) [PDF]

Think Right: Learning to Mitigate Under-Over Thinking via Adaptive, Attentive Compression
Joykirat Singh*, Justin Chih-Yao Chen, Archiki Prasad, Elias Stengel-Eskin, Akshay Nambi, Mohit Bansal* **(COLM 2026)** [PDF]

Agentic Reasoning and Tool Integration for LLMs via Reinforcement Learning
Joykirat Singh*, Raghav Magazine, Yash Pandya, and Akshay Nambi* (**NeurIPS FoRLM Workshop 2025**) [PDF]

Exposing the Achilles' Heel: Evaluating LLMs Ability to Handle Mistakes in Mathematical Reasoning
Joykirat Singh*, Akshay Nambi, Vibhav Vineet* (**ACL Main 2025**) [PDF]

PROMPTWIZARD*: Task-Aware Prompt optimization framework*
Eshaan Agarwal, ***Joykirat Singh****, Vivek Dani, Raghav Magazine, T. Ganu, Akshay Nambi* (**ACL Finding 2025**) [PDF]

Data-scarce Behavior Editing of Language Models
Joykirat Singh*, *Subhabrata Dutta*, T. Chakraborty* **(EMNLP Findings 2025)** [PDF]

How to think step-by-step: A mechanistic understanding of chain-of-thought reasoning
Subhabrata Dutta,* ***Joykirat Singh****, S. Chakrabarti, T. Chakraborty* - (**TMLR 2024**) [PDF]

Frugal LMs Trained to Invoke Symbolic Solvers Achieve Parameter-Efficient Arithmetic Reasoning
Subhabrata Dutta,* ***Joykirat Singh****, I. Pandey*, S. Manchanda, S. Chakrabarti, T. Chakraborty* (**AAAI 2024**) [PDF]

EROS: Entity-Driven Controlled Policy Document Summarization
Joykirat Singh*, Sehban Fazili*, Rohan Jain, Md. Shad Akhtar* - (**LREC-COLING 2024**) [PDF]

HONOR & AWARDS

- Awarded a silver medal for academic excellence at IIITD for achieving highest CGPA in computer science branch
- Awarded the Chanakya Fellowship by iHub Anubhuti IIITD (2022) to work on an independent research problem
- Presented the paper (top ~2%, Oral Presentation), Frugal LMs Trained to Invoke Symbolic Solvers Achieve Parameter-Efficient Arithmetic Reasoning at AAAI'24 Conference
- Volunteer, AAAI'24 Conference - received a travel grant to attend the conference

PROFESSIONAL SERVICES

Conference Reviewer
- COLM 2026
- NEURIPS 2026
- ACL 2025, EMNLP 2025 (ACL Rolling Review)
- ICLR 2024-25
- AAAI 2026

REFERENCES

- Mohit Bansal, John R. Louise S. Parker Professor of CS, UNC Chapel Hill.
- Sravan Bodapati, Sr. Staff Agentic AI SWE, Google
- Akshay Nambi, Principal Researcher, Microsoft Research India.
- Tanmoy Chakraborty, Associate Professor, IIT Delhi, India.
- Soumen Chakrabarti, Professor, IIT Bombay, India.